

Capital Bikeshare Demand Analysis and Forecasting

Overall Summary

This assessment uses the Capital Bikeshare hourly and daily rental dataset (2011 – 2012). The number of hourly records is 17379, whereas the daily records are 731. These records are used to characterize demand patterns, quantify the drivers of ridership statistically, and build a leakage-safe hourly forecasting pipeline under a custom monthly train/test partition.

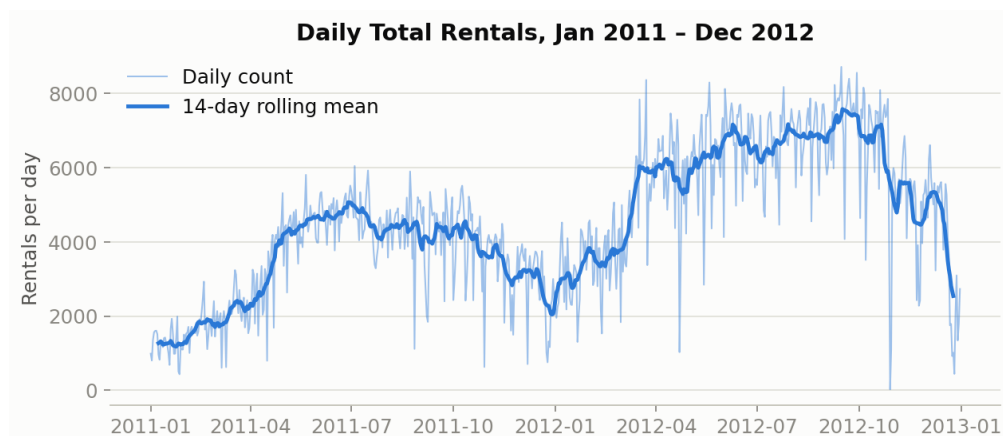
Initial Findings

The dataset provides two granularities that satisfy $cnt = casual + registered$ exactly with no conventional missing values, but *hour.csv* is not a complete hourly series. There are 165 missing hourly records (0.9%) spread across 76 of 731 days. Another finding was a full day of zero relative humidity readings (2011-03-10, 22 consecutive hours). Finally, the most severe weather category (Heavy Rain + Ice Pellets + Thunderstorm + Mist, Snow + Fog) is observed in only 3 of 17,379 hours, which is too rare to support a stable, generalizable coefficient on its own.

1. Task 1: Descriptive Statistics and Exploratory Analysis

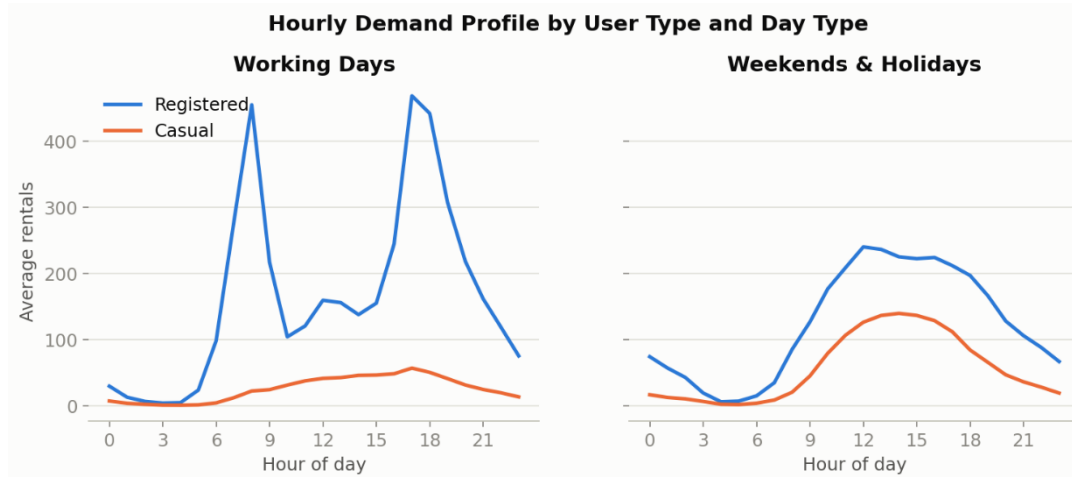
1.1. Growth and Seasonality

Total rentals grew from 1,243,103 (2011) to 2,049,576 (2012) — a 64.9% year-over-year increase, consistent with a maturing bikeshare network gaining stations and registered users. On the other hand, the seasonal cycle shows demand is lowest in spring (mean 2,604 rentals/day) and highest in fall (mean 5,644/day), with summer and winter in between.



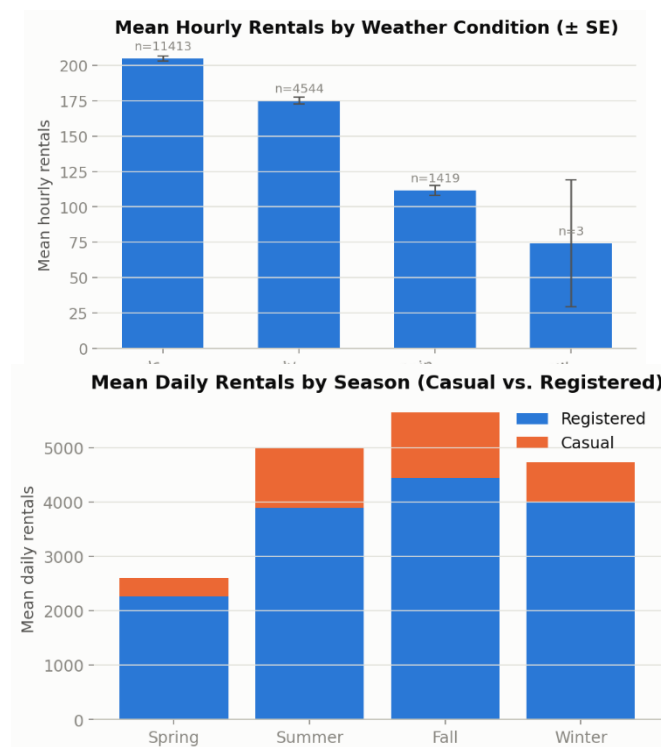
1.2. Commuters' vs Leisure Riders

Splitting the hourly profile by user type and day type produces the clearest signal in the dataset. On working days, registered ridership is sharply bimodal with a peak at 8 AM and a larger peak at 5–6 PM. On weekends, registered ridership is more spread out, with a peak around 12 PM and a secondary peak around 15 PM.



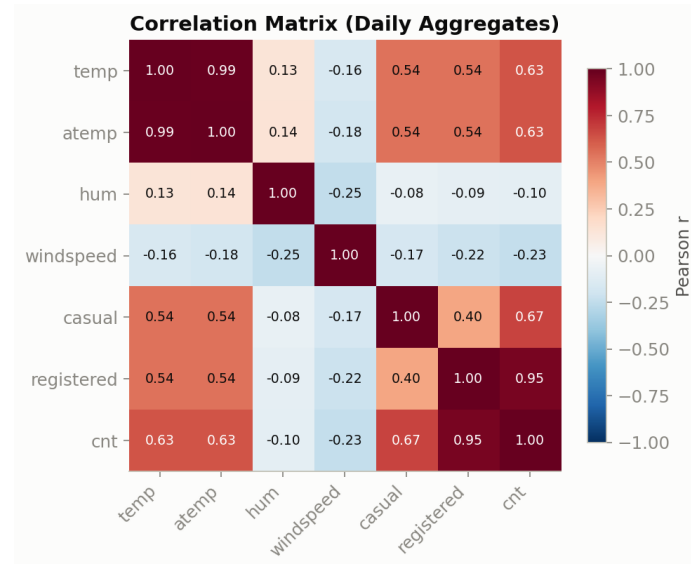
1.3. Weather and Seasonal Composition

Demand declines monotonically with worsening weather: mean hourly rentals fall from 204.9 (clear) to 175.2 (mist/cloudy) to 111.6 (light snow/rain) to 74.3 (severe — $n=3$, low confidence). Casual ridership is disproportionately concentrated in warm months. Its share of total volume more than doubles between the coldest and warmest weeks of the rolling 14-day series (roughly 6% to 27%), reinforcing that casual usage is discretionary and climate-sensitive while registered usage is comparatively stable year-round.



1.4. Correlation Matrix

Among continuous predictors, **temp** and **atemp** (feels-like temperature) correlate at $r = 0.99$, thus being effectively redundant. **temp** is the strongest single correlate of demand ($r = 0.63$ with **cnt**); humidity and windspeed correlate negatively but more weakly ($r = -0.10$ and -0.23 respectively).



1.5. Task 1 Summary

- Demand grew ~65% year-over-year, on top of a strong within-year seasonal cycle (trough: winter/spring; peak: fall) and a very strong within-day cycle.
- Registered riders (81% of volume) behave like commuters (sharp bimodal weekday peaks at 8 AM / 5–6 PM); casual riders (19%) behave like leisure users (single midday peak, far more weather- and temperature-sensitive, and their share of ridership rises in warm months / on weekends).
- Adverse weather suppresses demand monotonically (clear → mist → light precip → severe), roughly halving mean hourly rentals from clear to light-snow/rain conditions.
- Data-quality anomalies worth flagging for any downstream model: 165 missing hourly records (0.9%), one full day of a corrupted humidity sensor (2011-03-10), and a severe-weather category with only 3 observed hours in the entire two-year window.

2. Task 2: Statistical and Regression Analysis

2.1. Variable Selection Logic

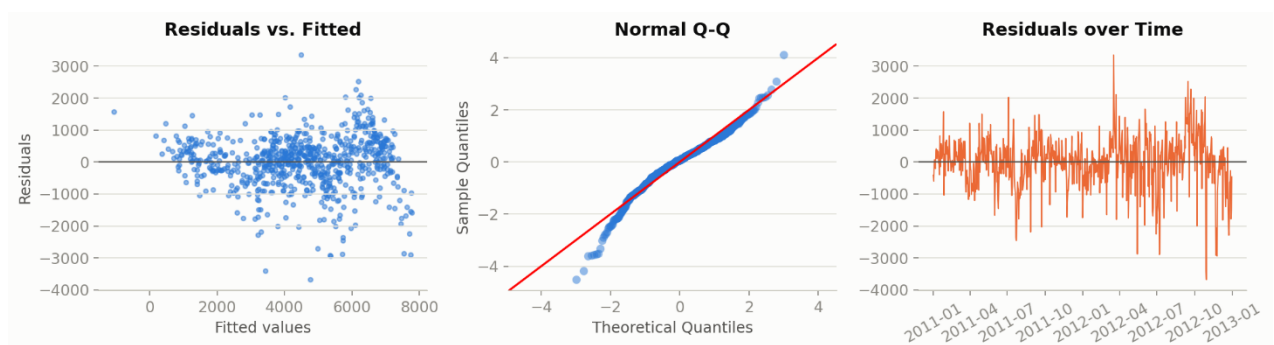
A variance-inflation-factor (VIF) screen confirms the correlation finding above: with both *temp* and *atemp* in the model, both have $VIF \approx 63$. Dropping *atemp* (retaining *temp* as the direct physical driver) leaves every remaining continuous/binary predictor at $VIF \approx 1.0\text{--}1.1$ — no material remaining multicollinearity. The working model regresses *cnt* (and, separately, *casual* and *registered*) on *season*, *year*, *holiday*, *working-day*, *weather condition*, *temperature*, *humidity*, and *windspeed*.

2.2. OLS Diagnostics

An OLS fit of the working formula explains a respectable 82.3% of daily variance ($R^2=0.823$), but three diagnostics disqualify it as the basis for formal statistical inference.

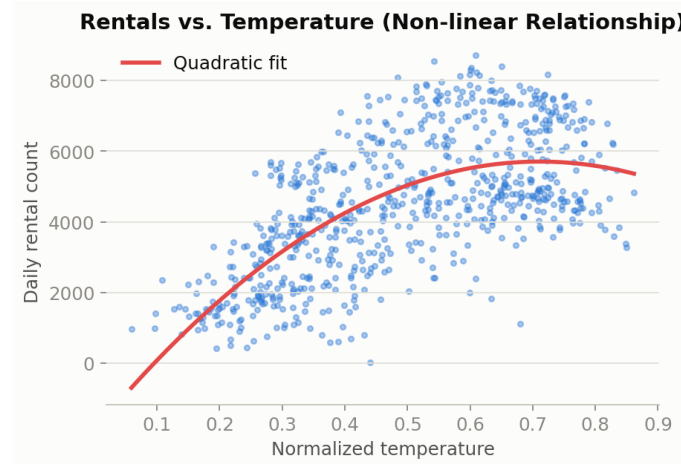
- **Breusch-Pagan test** rejects homoskedasticity ($LM=62.9$, $p<0.001$)
 - Heteroskedasticity: residual spread grows with the fitted level — expected for count data, where variance scales with the mean.
- **Jarque-Bera test** rejects normality ($JB=198.4$, $p<0.001$, skew= -0.64 , kurtosis= 5.21)
 - Non-normal residuals (left-skewed with heavy tails): a handful of days (extreme weather, holidays) are large negative outliers that a Gaussian-error model cannot represent well.
- **Durbin-Watson** statistic of 1.09 (vs. 2.0 for no autocorrelation)
 - Positive autocorrelation (well below the no-autocorrelation benchmark of 2.0): consecutive days' residuals are correlated which is unsurprising for a daily time series with unmodeled short-run persistence (weather streaks, local events)

Standard OLS errors are therefore unreliable for inference here without a HAC/Newey-West correction; OLS is retained only as a linear diagnostic benchmark, not as the basis for coefficient-level conclusions.



2.3. Non-linearity: Temperature effect

Adding a quadratic temperature term improves R^2 from 0.823 to 0.851 ($F=138.7$, $p<0.001$). A highly significant, substantively large improvement. Demand rises steeply with temperature to a comfort optimum around the mid-20s °C and then declines at the highest temperatures; a linear-only specification would systematically over-predict on the hottest days.



2.4. Count Data Distribution

The variance of daily *cnt* is approximately $833\times$ its mean. Here, Poisson's equidispersion assumption (variance = mean) is severely violated. A Negative Binomial GLM, which adds a free dispersion parameter is the standard and appropriate remedy.

2.5. Task 2 Summary

- After dropping *atemp* ($VIF \approx 63$ with temp), the remaining predictor set has no material multicollinearity.
- OLS residuals show heteroskedasticity, non-normality, and strong positive autocorrelation — standard symptoms of applying a Gaussian linear model to overdispersed daily count data
- We treat OLS as a diagnostic/benchmark only, not a basis for coefficient-level inference.
- Temperature's effect on demand is significantly non-linear (inverted-U): comfort-optimized, not monotonic.
- *cnt*'s variance is $\approx 833\times$ its mean; a Poisson fit is badly overdispersed (dispersion ≈ 172); a Negative Binomial GLM is the appropriate distributional choice and fits cleanly.

3. Task 3: Demand Forecasting and Predictive Modeling

Required partition (applied within every calendar month across the two-year span): train on days 1–20; test on day 21 through month-end. The target is total hourly rental count (*cnt*), predicted for every hour of every test window.

3.1. No Leakage Design

3.1.1. Repair Hourly Grid

The 165 missing hourly rows identified previously are reindexed onto a complete hourly index before any lag feature is built.

3.1.2. All autoregressive features use `.shift()` before `.rolling()`

This is a mechanical, unconditional guarantee that a feature computed for timestamp t only reads values with timestamp $< t$, regardless of whether that earlier timestamp happens to fall in a "train" or "test" month.

3.1.3. Model parameters are fit only on TRAIN rows

No statistic used at prediction time

3.1.4. Reading of “information available prior to the prediction period.”

We interpret this as: nothing with a later timestamp may be used (lookahead in the literal sense). A lag feature drawing on an already-elapsed, actually-observed value — even one a few days earlier within the same month's test window — is not lookahead; it is what a real deployed system would have in hand by that point. To keep this reading honest, an exogenous-only ablation (calendar+weather, no lag features at all) is reported alongside the full model so the lag features' marginal contribution is transparent rather than assumed.

3.2. Feature Engineering

Two feature families are constructed:

3.2.1. Calendar/Weather

season, year, holiday, working day, weather condition, temperature, humidity, windspeed, plus cyclical (sine/cosine) encodings of hour-of-day, day-of-week, and month

3.2.1. Autoregressive

lag_1h, lag_24h, lag_168h (one week), and rolling 24-hour/7-day means and a 24-hour rolling standard deviation, all built with the shift-then-roll discipline above.

3.3. Evaluation Metrics

Three metrics are reported:

3.3.1. RMSE

rentals/hr, primary: penalizes large misses, operationally relevant for fleet rebalancing.

3.3.2. MAE

rentals/hr, outlier-robust companion

3.3.3. MAPE

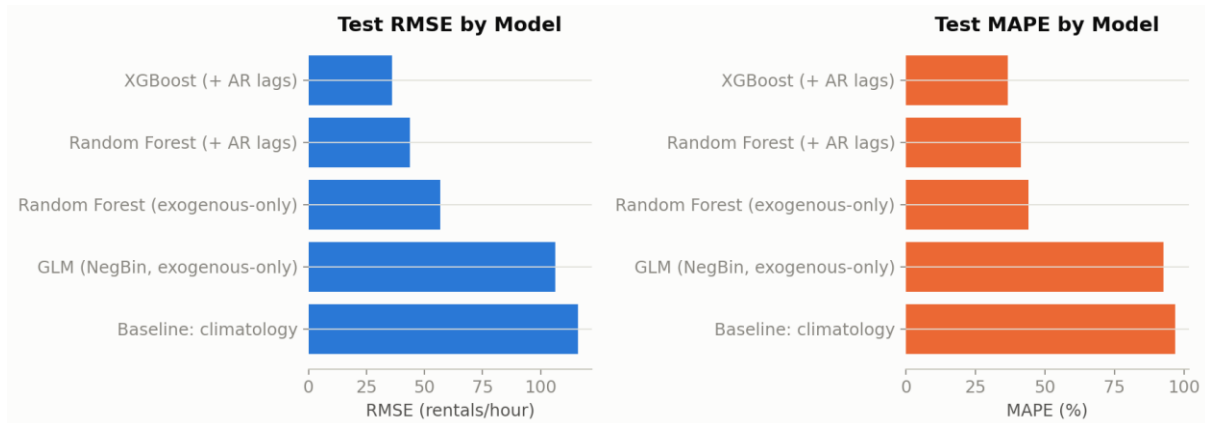
%, scale-free across months of different demand levels, but known to be unstable on bikeshare's many near-zero overnight hours. It is reported for completeness but weighted less heavily than RMSE/MAE in model selection.

3.4. Model Comparison

Five models are compared on the pooled 5,919-hour test set:

- Non-parametric hour $\times$ working-day climatology baseline
- Exogenous-only Negative Binomial GLM
- Random Forest fit on the same exogenous features only (isolating the effect of model class);
- Random Forest with the full feature set including AR lags
- XGBoost with the full feature set.

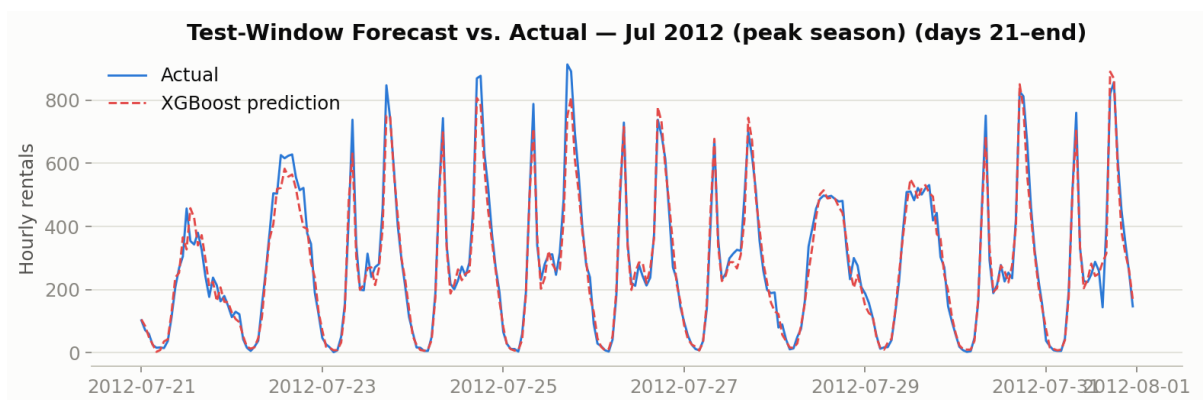
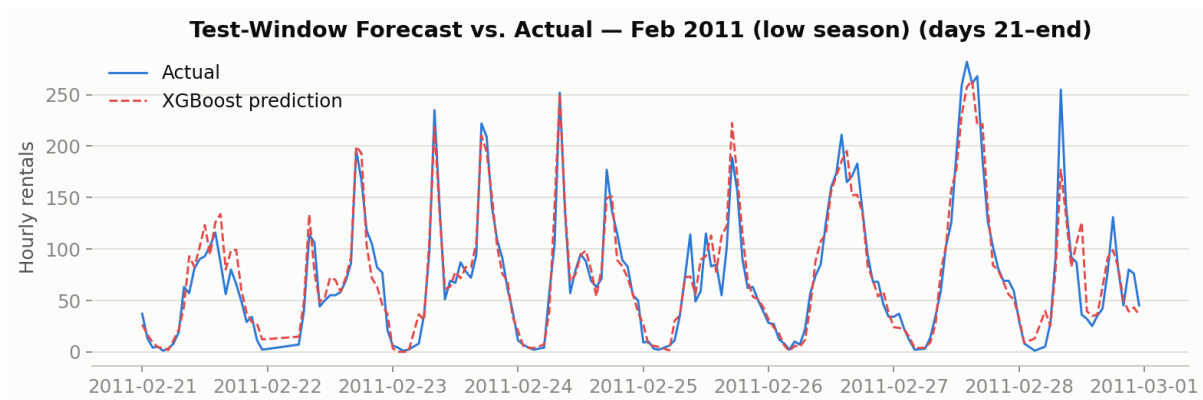
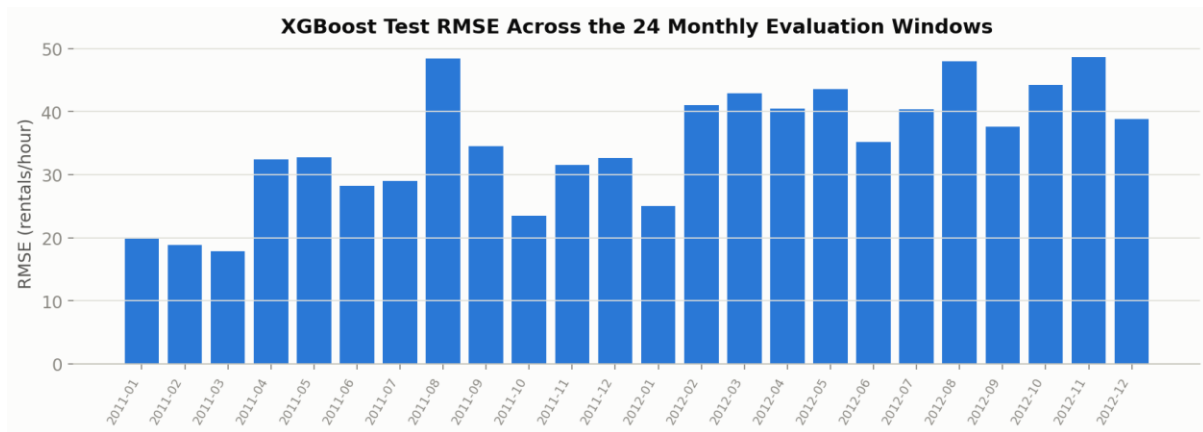
Model	RMSE	MAE	MAPE (%)	R2
Baseline: climatology	116.14	77.04	96.94	0.589
GLM (Negative Binomial, exogenous-only)	106.25	66.30	92.64	0.656
Random Forest (exogenous-only)	56.67	33.71	44.13	0.902
Random Forest (+ AR lags)	43.78	25.86	41.27	0.942
XGBoost (+ AR lags)	36.09	22.41	36.58	0.960



Two separable sources of improvement are visible. Switching from a log-linear GLM to a tree ensemble roughly halves error even with identical inputs (GLM 106.3 $\rightarrow$ RF-exogenous 56.7), because trees absorb the non-linear, bimodal time-of-day and inverted-U temperature effects without being told their functional form. Adding autoregressive history on top of that yields a further 20–36% reduction (RF-exogenous 56.7 $\rightarrow$ RF+lags 43.8 $\rightarrow$ XGBoost+lags 36.1), because recent actual demand is highly informative about the next few hours. XGBoost's regularized boosting edges out Random Forest at the same feature set and is the selected model.

3.5. Performance

Absolute RMSE trends upward from 2011 into 2012, but this tracks the 65% growth in demand level itself rather than a loss of model skill and is the reason MAPE is reported alongside RMSE. MAPE is noisiest in low-demand months (e.g., Aug-2011 and Dec-2012, where near-zero overnight hours inflate percentage error), not a sign of systematic failure, as the actual-vs-predicted traces confirm for both a low- and a high-season test window.



3.6. Task 3 Summary

- Built a leakage-safe hourly pipeline: gap-repaired grid $\rightarrow$ shift-then-roll AR features $\rightarrow$ train fit restricted to day-of-month ≤ 20 across all 24 months $\rightarrow$ predict on day-of-month ≥ 21 .
- Model class matters more than any single feature: moving from a log-linear GLM to tree ensembles roughly halves error at identical inputs, by absorbing the non-linear/bimodal structure documented in Tasks 1–2.
- Autoregressive history (1h/24h/168h lags, rolling means) provides a further, separable $\sim 20\text{--}36\%$ error reduction on top of the model-class gain.
- Best model: XGBoost with calendar + weather + AR-lag features — RMSE ≈ 36.1 rentals/hour, MAE ≈ 22.4 , MAPE $\approx 36.6\%$, $R^2 \approx 0.96$ on the pooled 5,919-hour test set spanning all 24 evaluation windows.

Conclusion

Across all three tasks, a consistent picture emerges. The Capital Bikeshare demand is driven by two behaviorally distinct populations

- commute-driven registered riders and
- weather/leisure-driven casual riders

Correctly modeling this system requires recognizing and testing for the count-data and non-linearity issues that a naive Gaussian linear model would miss, and an hourly forecasting pipeline that is explicit and defensible about what information is genuinely available at prediction time.